

Topics Covered

- Confusion Matrix
- Accuracy
- Precision & Recall
- F1-Score
- TPR, TNR, FPR & FNR
- ROC Curve
- AUC
- Python Implementation

Confusion Matrix

❑ A **Confusion Matrix** is a table used to evaluate the performance of a classification model by comparing actual and predicted class labels.

❑ Components of a Confusion Matrix:

The confusion matrix is organized into four parts:

- **True Positive (TP):** The number of instances where the model correctly predicted the positive class.
- **True Negative (TN):** The number of instances where the model correctly predicted the negative class.
- **False Positive (FP):** The number of instances where the model incorrectly predicted the positive class (also known as a Type I error).
- **False Negative (FN):** The number of instances where the model incorrectly predicted the negative class (also known as a Type II error).

Confusion Matrix for Binary Classification

Table 1: *Confusion Matrix for Binary Classification*

		Predicted Label	
		0 (False)	1 (True)
Actual Label	0 (False)	TN	FP
	1 (True)	FN	TP

- Positive class (1) = True
- Negative class (0) = False
- Rows = Actual labels
- Columns = Predicted labels
- Diagonal (TP, TN) = Correct Predictions
- Off-Diagonal (FP, FN) = Incorrect Predictions
- The confusion matrix forms the basis for Accuracy, Precision, Recall and F1-Score.

Confusion Matrix for Binary Classification

Table 2: *Confusion Matrix for Email Classification*

		Predicted Label	
		0 (False)	1 (True)
Actual Label	0 (False)	45 (TN)	10 (FP)
	1 (True)	5 (FN)	40 (TP)

- Suppose we have a binary classification problem where we predict whether an email is spam (positive) or not spam (negative). Let's assume the model made predictions for 100 emails, and the following results were obtained:
- **True Positives (TP):** 40 (emails that are correctly classified as spam)
- **False Positives (FP):** 10 (emails that are incorrectly classified as spam)
- **True Negatives (TN):** 45 (emails that are correctly classified as not spam)
- **False Negatives (FN):** 5 (emails that are incorrectly classified as not spam)

Accuracy Performance Metric

- ❑ Accuracy measures the proportion of correctly classified instances out of the total number of instances.

$$\textit{Accuracy} = \frac{TP + TN}{TP + FP + TN + FN}$$

- Accuracy calculation for Email classification confusion matrix
- $TP = 40, TN = 45, FP = 10, FN = 5$
- Accuracy

$$\textit{Accuracy} = \frac{40 + 45}{40 + 10 + 45 + 5} = \frac{85}{100} = 0.85$$

Recall & Precision

- ❑ Recall measures the proportion of actual positive instances that are correctly identified by the model.

$$\text{Recall} = \frac{TP}{TP + FN}$$

$$\text{Recall} = \frac{40}{40 + 5} = \frac{40}{45} = 0.89$$

- ❑ Precision measures the proportion of predicted positive instances that are actually positive.

$$\text{Precision} = \frac{TP}{TP + FP}$$

$$\text{Precision} = \frac{40}{40 + 10} = \frac{40}{50} = 0.80$$

F1-Score Performance Metric

- ❑ F1-Score is the harmonic mean of Precision and Recall, providing a balanced measure of model performance.

$$F1 - Score = \frac{2 * Precision * Recall}{Precision + Recall}$$

$$F1 - Score = \frac{2 * 0.80 * 0.89}{0.80 + 0.89} = \frac{2 * 0.712}{1.69} = 0.84$$

F1-Score balances both Precision and Recall, making it useful for imbalanced datasets.

TPR, TNR, FPR, FNR

- ❑ True Positive Rate (TPR) / Recall / Sensitivity: Out of all actual positive cases, how many were correctly identified?

$$TPR = \frac{TP}{TP + FN}$$

- ❑ False Negative Rate (FNR) : Out of all actual positive cases, how many were missed?

$$FNR = \frac{FN}{TP + FN}$$

- ❑ True Negative Rate (TNR) / Specificity : Out of all actual negative cases, how many were correctly identified?

$$TNR = \frac{TN}{TN + FP}$$

- ❑ False Positive Rate (FPR) : Out of all actual negative cases, how many resulted in a false alarm?

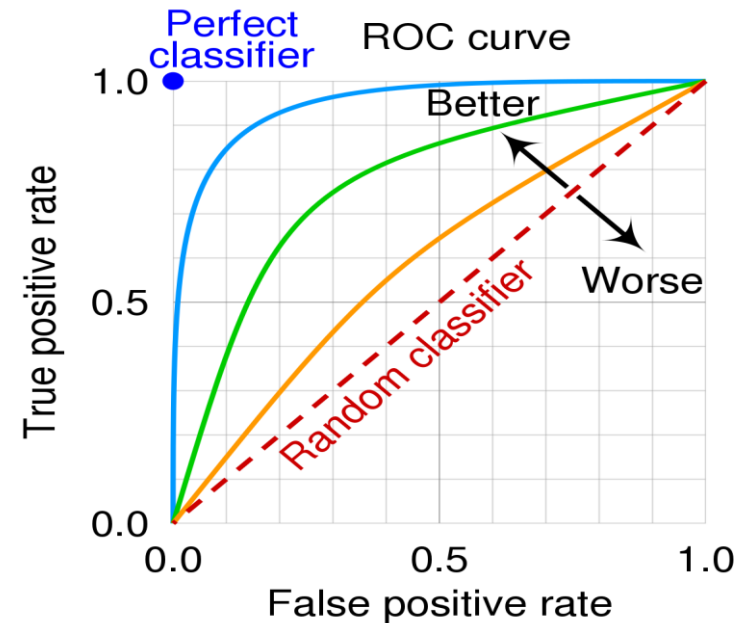
$$FPR = \frac{FP}{TN + FP}$$

- ❑ $TPR + FNR = 1$
- ❑ $TNR + FPR = 1$

Receiver Operating Characteristic Curve

❑ The ROC Curve is a graphical representation of a classifier's performance by plotting the True Positive Rate (TPR) against the False Positive Rate (FPR) at different classification thresholds.

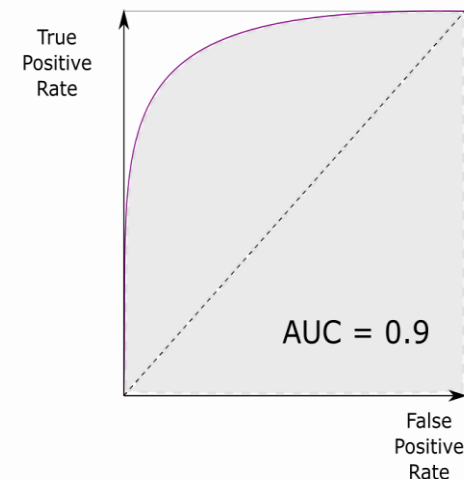
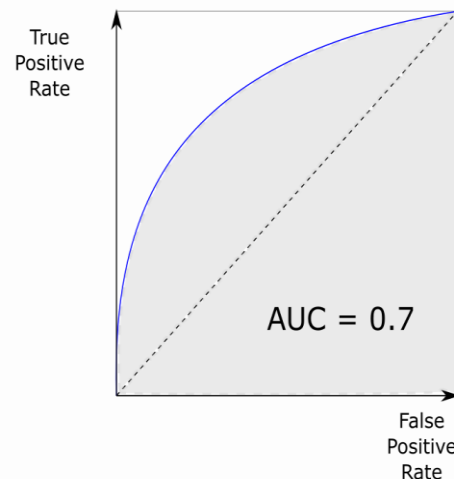
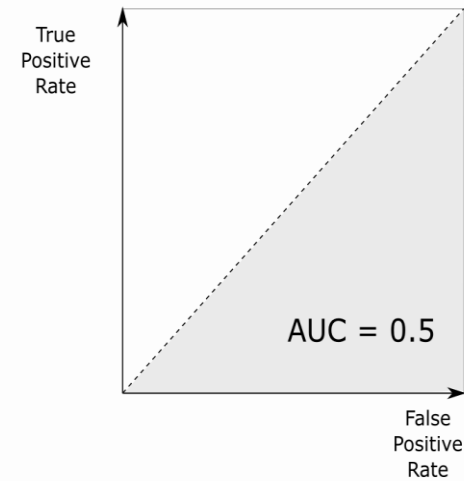
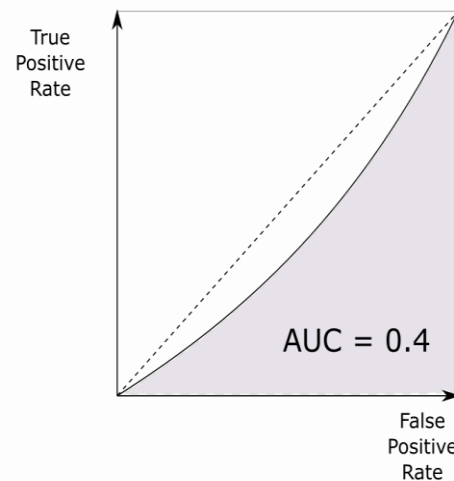
- X-axis : False Positive Rate (FPR)
- Y-axis : True Positive Rate (TPR)
- Each point represents a different classification threshold.
- A curve closer to the top-left corner indicates better performance.



AUC (Area Under the ROC Curve)

❑ AUC measures the overall ability of a classifier to distinguish between positive and negative classes.

- AUC values range from 0 to 1
- **AUC = 1**: Perfect classifier
- **AUC = 0.9** : Excellent classifier
- **AUC = 0.7**: Good classifier.
- **AUC = 0.5**: Random guessing
- **AUC = 0.4** : Poor Classifier



Performance Metrics in Python

```
4 @author: kajal
5 """
6 from sklearn.metrics import (
7     confusion_matrix,
8     accuracy_score,
9     precision_score,
10    recall_score,
11    f1_score
12 )
13
14 # Actual labels
15 y_true = [1, 1, 1, 1, 1, 0, 0, 0, 0, 0]
16
17 # Predicted labels
18 y_pred = [1, 1, 0, 1, 1, 0, 0, 1, 0, 0]
19
20 # Performance Metrics
21 cm = confusion_matrix(y_true, y_pred)
22 accuracy = accuracy_score(y_true, y_pred)
23 precision = precision_score(y_true, y_pred)
24 recall = recall_score(y_true, y_pred)
25 f1 = f1_score(y_true, y_pred)
26
27 # Print Results
28 print("Confusion Matrix:")
29 print(cm)
30
31 print("\nAccuracy :", round(accuracy, 2))
32 print("Precision:", round(precision, 2))
33 print("Recall   :", round(recall, 2))
34 print("F1-Score :", round(f1, 2))
35
```

Run plot-generating code in the Editor or IPython console to see your figures. JupyterLab only supports static images, so it can't display interactive plots like Bokeh.

Help Variable Explorer Debugger Profiler Plots File

Console 1/A x

```
In [83]: %runfile C:/Users/kajal/basic_cm.py --wdir
Confusion Matrix:
[[4 1]
 [1 4]]

Accuracy : 0.8
Precision: 0.8
Recall   : 0.8
F1-Score : 0.8

In [84]: |
```